

Aakarsh Saxena

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EDUCATION

Indian Institute of Information Technology Lucknow

B.Tech in Information Technology

Aug 2024 – June 2028

Lucknow, Uttar Pradesh

Atomic Energy Central School No. 4

Class XII: 91.8% | Class X: 88.4%

2022 – 2024

Rawatbhata, Rajasthan

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, JavaScript, SQL, Bash

Core CS: Data Structures & Algorithms (500+ solved), OOP, Operating Systems, Computer Networks, DBMS, System Design, Distributed Systems, Probability & Statistics

Backend & Systems: Node.js, Express.js, FastAPI, Next.js, REST APIs, JWT & HMAC Auth, Docker, Linux/Unix, Git, CI/CD (GitHub Actions)

Data & Databases: PostgreSQL, MongoDB, Redis, SQLite, Drizzle ORM, Schema Design, Idempotency & Transactional Outbox, A/B Testing & Causal Inference

AI/ML: Machine Learning, LLM Agent Orchestration, Reinforcement Learning, GRPO, LoRA Fine-Tuning, Model Evaluation, Prompt Engineering, Gemini API, Qwen2

Frontend & Testing: React 19, Vite, TailwindCSS, Three.js, pytest, Property-Based Testing, mypy -strict, Zod

ACHIEVEMENTS

LeetCode Knight | **Codeforces Specialist** | **CodeChef 3★** — solved **500+** DSA problems across LeetCode, CodeChef, and Codeforces

Global Rank 4 at CodeChef Starters 233 (Rated); **Global Rank 69** (Biweekly 188) and **Rank 143** (Weekly 506) among **30,000+** participants in LeetCode rated contests

Rank 1200 among **90,000+** registrations in **Amazon ML Challenge** (team) with **97.8% accuracy**

Finalist at **Meta × Scaler OpenEnv AI Hackathon** — among top teams building production-grade AI systems at scale

Qualified for Round 3 of **Flipkart GRiD 6.0**, a national-level engineering and problem-solving competition

PROJECTS

RevPilot AI: Autonomous Revenue-Recovery & Attribution Agent

Razorpay AI Buildathon 2026

Python, FastAPI, Next.js, SQLite, LLM Agents, Statistical Inference

[GitHub](#)

- Engineered a 7-stage autonomous recovery agent (detect → diagnose → propose → policy → execute → verify → attribute) over Razorpay payment telemetry; a deterministic rule engine resolves **159/199** diagnoses at **96.4%** accuracy, beating the LLM's 90.4% on a golden set, with only ambiguous failures routed to inference.
- Stripped the LLM of all authority over money via a deterministic policy firewall and capability-token executor (consent, amount caps, DND, quiet hours, 12 stopping rules); an ablation across **5 decision policies** showed removing it caused **284 policy breaches** at identical recovery, with termination and token isolation proved by property tests across **1,282 tests** under mypy -strict.
- Attributed *incremental* rather than gross recovery through a randomized holdout — crediting **INR 0** to 17 organic conversions over a reproducible 420-transaction corpus — and verified two live Razorpay Test-Mode payments via HMAC-signed webhooks plus an API reconciler for lost deliveries.

Aegis: AI-Powered Code Analysis & Auto-Fix Platform

2026

Next.js 15, TypeScript, PostgreSQL, Drizzle ORM, Gemini 2.5

[GitHub](#)

[Live Demo](#)

- Built a serverless, multi-tenant full-stack platform on Next.js 15 and TypeScript, deployed on Vercel, with secure authentication and isolated per-user workspaces for connected GitHub repositories.
- Developed an AI Scanner that analyzes connected repos via Gemini 2.5 Flash, detects bugs and code issues, and auto-opens pull requests with generated patches, cutting manual bug-triage effort by an estimated **60%**.
- Designed a Drizzle ORM schema over Neon serverless PostgreSQL with strict per-tenant query scoping and AES-256-GCM encryption for stored API keys, supporting concurrent multi-user workspaces with zero data leakage.

Oceanus: Multi-Agent RL Environment for Policy Negotiation

Meta × Scaler Finalist

Python, FastAPI, Qwen2, GRPO, LoRA, Three.js

[GitHub](#)

[Live Demo](#)

- Architected a distributed 3-layer reinforcement learning system (swarm simulation, policy negotiation, adversarial curriculum generation) in which decoupled, independently scalable LLM agents coordinate resource recovery via natural-language negotiation.
- Fine-tuned Qwen2-0.5B with GRPO and LoRA adapters on custom multi-agent reward signals, achieving a **+9.0 average reward improvement** over baseline policies across progressively harder adversarial curriculum stages.
- Built concurrent FastAPI inference pipelines supporting parallel multi-agent rollouts, plus interactive Three.js visualizations for analyzing emergent coordination behavior across **100+** simulated negotiation episodes.